

Subtitle Basic idea about DC to AC converter
Institute Dr B C Roy Engineering College, Durgapur
Subject Power Electronics PC EE 504

Inverter Lecture 04

Kingsuk Majumdar

Hello friends

Lecture 04

- ① Brief recap of last lecture 03
- ② A brief review of PWM
- ③ Single pulse width modulation Inverter
- ④ Signification
- ⑤ References

A brief review of PWM

- PWM:

Pulse width modulation (PWM), or pulse-duration modulation (PDM), is a method of reducing the average power delivered by an electrical signal, by effectively chopping it up into discrete parts. The average value of voltage (and current) fed to the load is controlled by turning the switch between supply and load on and off at a fast rate. The longer the switch is on compared to the off periods, the higher the total power supplied to the load. [<https://en.wikipedia.org/>]

NB:

1. <https://www.youtube.com/watch?v=YmPziPfaByw>
2. https://www.multisim.com/content/DYJKviXg8EJwKGPvzPRXv5/pwmbyic555_km/open/

PWM inverter

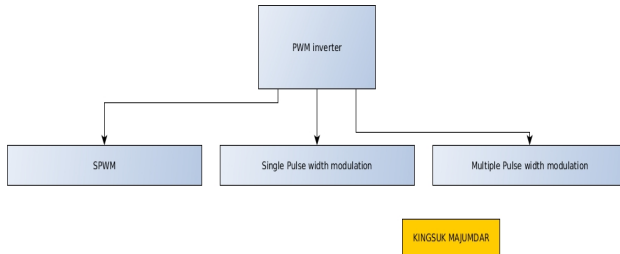


Figure: PWM inverter type

Single pulse width modulation Inverter

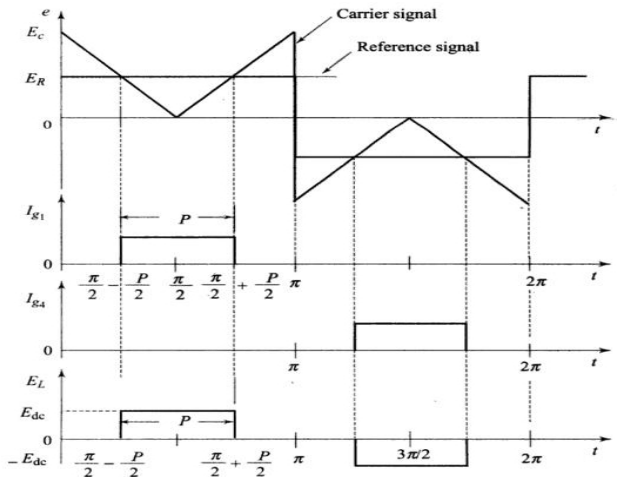


Figure: Single pulse width modulation Inverter

NB:picture taken from google

AM index and FM index

- AM index (Amplitude modulation index)

$$M_a = \frac{A_{ref}}{A_{car}} \quad (1)$$

- FM index (Frequency modulation index)

$$M_f = \frac{f_{car}}{f_{out}} \quad (2)$$

Circuit for Single pulse width modulation Inverter

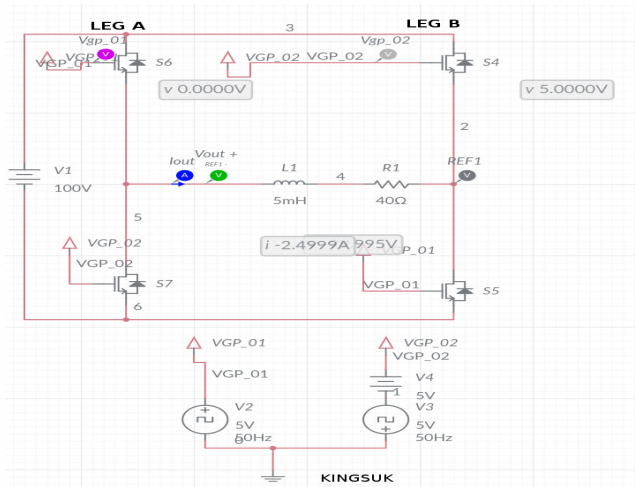


Figure: Circuit for Single PWM inverter

Single PWM mathematics

As the output waveform has quasi-square symmetry, hence, DC and cosine terms in the Fourier's series will be zero. take $\delta = 2 * d$ pulse width of the output voltage waveform.

$$b_n = \frac{2}{\pi} \int_{\frac{\pi}{2}-d}^{\frac{\pi}{2}+d} V_s \sin(n\omega t) = \frac{4V_s}{n\pi} [\sin(\frac{n\pi}{2}) \sin(nd)] \quad (3)$$

Hence, output voltage

$$V_{oave} = \sum_{n=1,2,3...}^{\inf} \frac{4V_s}{n\pi} [\sin(\frac{n\pi}{2}) \sin(nd)] \sin(n\omega t) \quad (4)$$

cont...

- $\sin(\frac{n\pi}{2}) \Rightarrow$ this factor decides that n^{th} harmonic will exist or **NOT** and (if exist) what would be its sign i.e. +ve or -ve.
- $[\sin(\frac{n\pi}{2})\sin(nd)] \Rightarrow$ Responsible for amplitude of the n^{th} harmonic.
- $\sin(nd) \Rightarrow$ use to mitigate a particular harmonic

Therefore,

$$V_{oave} = \frac{4V_s}{\pi} [\sin(d) \sin(\omega t) - \frac{1}{3} \sin(3d) \sin(3\omega t) + \frac{1}{5} \sin(5d) \sin(5\omega t) - \dots] \quad (5)$$

Signification:

- pulse width ($\delta = 2 * d = \pi \text{rad}$) from eq:(4), the fundamental component of voltage wave becomes maximum

$$V_{o1max} = \frac{4V_s}{\pi} \quad (6)$$

- $\delta \neq \pi \text{rad}$

$$V_{o1m} = \frac{4V_s}{\pi} \sin(d) \quad (7)$$

- mitigate a particular harmonic say 3^{rd} harmonic take $d = \frac{\pi}{n}$ i.e.

$$\delta = \frac{2\pi}{n} \text{ here } n=3$$

$$\text{hence } \delta = \frac{2\pi}{3}; \delta = 120^\circ; d = 60^\circ$$

MATLAB file exchange (for this type of mitigation): <https://in.mathworks.com/>

[matlabcentral/fileexchange/56474-fourier-analysis-of-a-single-pwm-signal-a-particular-harmonic-mitigation](https://in.mathworks.com/matlabcentral/fileexchange/56474-fourier-analysis-of-a-single-pwm-signal-a-particular-harmonic-mitigation)

cont..

The magnitude of 3rd harmonic

$$V_{o3m} = \frac{4V_s}{3\pi} \sin(3d) \quad (8)$$

$$V_{o3m} = \frac{4V_s}{3\pi} \sin(3 * 60); d = 60^\circ \quad (9)$$

$$V_{o3m} = 0 \quad (10)$$

But it is also affected the fundamental i.e.

$$V_{o1m} |_{d=60^\circ} = \frac{4V_s}{\pi} \sin(60) \quad (11)$$

$$V_{o1m} |_{d=60^\circ} = 0.5513 V_s \quad (12)$$

References



N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics. Converters, Applications and Design*, 3rd ed. John Wiley and Sons, Inc, 2003.



M. H. Rashid, *Power electronics handbook: devices, circuits and applications*. Elsevier, 2010.



P. Bimbhra and S. Kaur, *Power electronics*. Khanna publishers, 2012, vol. 2.